09 — risc8.xdc Explained: The Vivado Constraints File

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1. Overview

`risc8.xdc` is not Verilog. It is a **constraints file** written in Tcl, the command language of Vivado. It tells Vivado how fast the clock should run, so the timing report after implementation can say whether the design is fast enough. Later it can also hold the board pin numbers.

Item	Value
File	constraints/risc8.xdc (19 lines)
Language	XDC (Xilinx Design Constraints), based on Tcl
Added in Vivado as	Constraints
Applies to	The design top, risc8_5stage_core (its port `clk`)
Calls	One Vivado command: create_clock, and get_ports inside it
Used by	Vivado synthesis and implementation (ignored by simulation)

1.1 Commands used and where they belong

Command	What it is	Belongs to	Purpose here
create_clock	Vivado Tcl command	Vivado / XDC standard	Declares the clock and its period
get_ports	Vivado Tcl command	Vivado / XDC standard	Finds the port named clk on the top module
set_property (only in comments)	Vivado Tcl command	Vivado / XDC standard	Example of how to assign board pins later

2. Line-by-line walkthrough

2.1 Header comments (lines 1–17)

```
## ============================================================================
##  FILE    : risc8.xdc   (Vivado constraints file)
##  PROJECT : 8-bit RISC Processor Core with 5-Stage Pipeline
## ----------------------------------------------------------------------------
##  WHAT THIS FILE DOES
##    Tells Vivado the clock speed we want, so the timing report after
##    implementation shows whether the design is fast enough.
##    10.000 ns period = 100 MHz target clock.
##
##  PIN ASSIGNMENTS
##    None are given, because they depend on your FPGA board.
##    Synthesis, implementation and timing work without them.
##    Only "Generate Bitstream" needs pins. To program a board, add
##    lines like these with the pin names from your board's manual:
##      set_property PACKAGE_PIN W5 [get_ports clk]
##      set_property IOSTANDARD LVCMOS33 [get_ports clk]
## ============================================================================
```

In XDC, a line starting with `#` is a comment (the file uses `##` for clarity). The header explains the target clock and why no pins are given yet. It also shows the two lines you would add to connect `clk` to a board pin.

2.2 The clock constraint (line 19)

```
create_clock -period 10.000 -name sys_clk [get_ports clk]
```

Part	Meaning
create_clock	Tell Vivado that a signal is a clock
-period 10.000	One clock cycle lasts 10.000 ns, so the target is 100 MHz
-name sys_clk	A name for this clock, used in the timing reports
[get_ports clk]	Apply it to the input port called clk of the top module, risc8_5stage_core.v

After implementation, **Report Timing Summary** checks every path between flip-flops against 10 ns. If **WNS** (worst negative slack) is positive, the design meets 100 MHz.

3. Adding pins for a real board (optional)

To generate a bitstream, every port needs a pin and a voltage standard. Example for a Basys-3 board (check your own board's manual):

```
set_property PACKAGE_PIN W5  [get_ports clk]
set_property IOSTANDARD LVCMOS33 [get_ports clk]
set_property PACKAGE_PIN U18 [get_ports rst]
set_property IOSTANDARD LVCMOS33 [get_ports rst]
set_property PACKAGE_PIN U16 [get_ports halted]
set_property IOSTANDARD LVCMOS33 [get_ports halted]
```

Port	Connect to
clk	Board oscillator
rst	A push button
halted, pc_out[7:0], flags_out[1:0], instr_done	LEDs

On a board the program finishes in about half a microsecond, too fast to see. To watch it on LEDs, slow the clock with a clock divider (for example to 1 Hz).

4. Common mistakes

- Writing the port name differently from the Verilog (it must be exactly `clk`).
- Adding the .xdc file as a design source instead of a constraint.
- Generating a bitstream without pin assignments: Vivado stops with an "unconstrained ports" error.
- Using pin numbers from a different board.

5. Viva questions

Q1. What is an XDC file? A Xilinx Design Constraints file: Tcl commands that tell Vivado about clocks, timing and pins. It is not part of the Verilog design.

Q2. What does create_clock -period 10.000 mean? The clock period is 10 ns, so the target frequency is 100 MHz.

Q3. Why is there no pin assignment? Pins depend on the board. Simulation, synthesis, implementation and timing analysis do not need them; only the bitstream does.

Q4. How do you know the design meets the clock? After implementation, the timing summary shows WNS (worst negative slack). It must be zero or positive.